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Graded Assignment • 20 min

Due Mar 13, 11:59 PM CET

1. What are some problems of ground-based telescopes? (Pick all that apply) — 1 point

- [ ] Ground-based telescopes can only see objects in our own Galaxy.
- [x] The atmosphere distorts light, blurring images.
- [ ] Telescopes on Earth are too expensive compared to space-based telescopes.
- [x] The Earth's atmosphere absorbs some wavelengths of light.

2. Which one of the following is NOT a difficulty when operating a telescope in space? — 1 point

- (●) It is too hard to analyze X-ray data.
- ( ) The heat and light from the sun must be blocked.
- ( ) A telescope in Low Earth Orbit has difficulty transmitting data back to one receiving station on Earth.
- ( ) Telescopes in space must be able to withstand collisions with human debris and natural dust particles.

3. One solution to transmitting data from a telescope in space to the Earth is by using a satellite in geostationary orbit. A satellite in this orbit stays directly above the same spot on the ground, and thus goes around the earth in exactly one day. For an object to be in a geostationary orbit, it must orbit at an altitude of 5.6 Earth radii above the surface of the Earth (6.6 Earth radii from the Earth's center). — 1 point

A geostationary orbit can be above a point on only one great circle of the Earth. Which is it?

- ( ) The Prime Meridian.
- ( ) The International Date Line.
- ( ) The Tropic of Capricorn.
- (●) The Equator

4. Cygnus X1 was one of the earliest black holes discovered. From the X-ray variability, the diameter of the accretion disk is known to be no larger than $3000\ \mathrm{km}$. From the orbital dynamics of the companion blue supergiant, the mass inside the disk is roughly 16 times the mass of the Sun (note that the mass of the Sun is roughly 2×10^{33} g). — 1 point

If all that mass were evenly distributed in a sphere of a diameter of $3000\ \mathrm{km}$, what would be the average density of Cygnus X1?

- ( ) $2.5 \times 10^{30}\ \mathrm{g/cm^3}$
- ( ) $2.5 \times 10^{25}\ \mathrm{g/cm^3}$
- (●) $2.2 \times 10^{9}\ \mathrm{g/cm^3}$
- ( ) $2.8 \times 10^{23}\ \mathrm{g/cm^3}$

5. The active galaxy 3C279 is very bright and variable in the X-rays due to emission from gas that is falling into a central super-massive black hole. Based on the X-ray variability, the size of the central emission region is estimated to be no larger than 7.8 times the size of our solar system. What is the fastest timescale this entire emission region can vary in brightness? (Use 10^{13} meters for the solar system diameter.) — 1 point

- ( ) 0.0025 days
- ( ) 260,000 days
- ( ) 467 days
- (●) 3 days